FINAL EXPLANATION: CHEMISTRY GRADE 10

Student Assessment Document

FINAL EXPLANATION: CHEMISTRY | GRADE 10 | SUB-STRAND 1.5: PERIODICITY	
Student Name	______________________________
Class	______________________________
Date	______________________________

(The table's second header row reads: **Student Assessment Document**)

INSTRUCTIONS FOR STUDENTS

This document is your Final Explanation guide. Your goal is to write a complete, well-reasoned answer to the driving question using everything you have learned across the seven lessons of this unit. Read the driving question carefully, then use each section below as a building block for your full answer. For each section, a prompt tells you what to address, and an exemplar shows you the quality and style expected. Write in your own words — do not copy the exemplar. Your final answer must connect back to the anchoring phenomenon: the exploding metals demonstration at the Kisumu Water and Sanitation Company. A strong Final Explanation tells a coherent scientific story from protons and electrons all the way to the orange flame skimming across Lake Victoria water.

Section 1: Atomic Structure and Electron Configuration — Building the Foundation	
Explain what an atom is made of and how electrons are arranged in shells around the nucleus. Define atomic number, electron shells, and valence electrons. Use sodium (Na, atomic number 11) and magnesium (Mg, atomic number 12) as your main examples. Why does one extra proton — going from Na to Mg — change how electrons are arranged, and why does that matter?	Every element on the periodic table is defined by its atomic number — the number of protons in its nucleus. Sodium has 11 protons, so its atomic number is 11. Its 11 electrons are arranged in three shells: 2 in the first shell, 8 in the second, and 1 lone electron in the outermost (valence) shell. Magnesium has 12 protons and 12 electrons, arranged as 2, 8, 2 — giving it two valence electrons. This single-proton difference is the key to everything. The extra proton in magnesium's nucleus pulls more strongly on those outer electrons, holding them more tightly in place. Sodium's single valence electron, sitting relatively far from the nucleus and weakly held, is much easier to remove. This difference in electron arrangement — specifically in the number and hold of valence electrons — is the foundation for explaining why sodium explodes in Lake Victoria water while magnesium barely fizzes.

Section 2: Ionisation Energy, Shielding, and Periodic Trends — The Forces at Work	
Define ionisation energy and	Ionisation energy is the energy needed to remove an electron from an atom in its gaseous state. The lower an element's ionisation

electron shielding. Explain how and why ionisation energy changes (a) across Period 3 from left to right (e.g., Na → Mg → Al → Cl → Ar) and (b) down Group I from lithium to sodium to potassium. Use the concepts of nuclear charge, shielding by inner electron shells, and distance from the nucleus. Connect these trends to the reactivity gradient you observed in the Kisumu demonstration.	energy, the more easily it loses an electron — and for metals, losing electrons easily means reacting vigorously. Two factors control ionisation energy: nuclear charge (how many protons are pulling inward) and shielding (how much the inner electron shells block that pull from reaching the outer electrons). Across Period 3, as you move from sodium to magnesium to aluminium and beyond, the nuclear charge increases with each step. The number of inner shielding electrons stays roughly the same, so the effective nuclear charge felt by the valence electrons increases. This makes it progressively harder to remove a valence electron, and reactivity with water decreases. That is exactly what the water purification officer demonstrated: sodium exploded, magnesium barely fizzed, and aluminium did nothing in cold water. Going down Group I, the trend reverses. Lithium, sodium, and potassium all have one valence electron, but each successive element adds a new full electron shell between the nucleus and that outer electron. This dramatically increases shielding and increases the distance of the valence electron from the nucleus. The effective nuclear charge felt by the outer electron drops, ionisation energy falls, and the element becomes more reactive — which is why potassium's explosion in the demonstration was more violent than sodium's.

Section 3: Explaining the Phenomenon — From Electron Configuration to Orange Flame

Now connect your scientific understanding directly to the anchoring phenomenon. Explain at a particle level why sodium reacts violently with water, why magnesium reacts only slightly, and why argon does not react at all. Your explanation must reference valence electrons, ionisation energy, electron configuration, and the products of the sodium–water reaction (including why the solution becomes alkaline and why there is a flame). Why is argon, sitting right next to the highly reactive chlorine, completely inert?	When sodium is dropped into Lake Victoria water, its single, loosely held valence electron is transferred rapidly to water molecules. This is a redox reaction: sodium loses its valence electron (it is oxidised), and water molecules accept electrons to release hydrogen gas (H_2) and hydroxide ions (OH^-). The hydrogen gas ignites immediately because the reaction releases enough heat — the strongly alkaline solution left behind is sodium hydroxide (NaOH), which turns red litmus paper blue. The reaction is so vigorous because sodium's first ionisation energy is low: only 496 kJ/mol. Its one valence electron is shielded by two full inner shells and is relatively far from the nucleus. Magnesium has two valence electrons and a higher nuclear charge (12 protons vs. 11). Its first ionisation energy is 738 kJ/mol, and its second is much higher still — removing both valence electrons to react with cold water requires more energy than cold water can supply, so only a slow surface reaction occurs. Aluminium, with three valence electrons and 13 protons, has an even higher effective nuclear charge and also forms a protective oxide layer, so it shows no visible reaction. Argon's complete inertness is explained by its electron configuration: 2, 8, 8 — a full outer shell. With no vacancy and no tendency to gain or lose electrons, its ionisation energy is extremely high (1521 kJ/mol) and it forms no bonds. Chlorine, its neighbour, has seven valence electrons and desperately needs one more to complete its shell, making it highly reactive — but argon is already 'satisfied.' Position in the periodic table therefore tells us, with precision, how an element will behave before we ever put it in a test tube.

Section 4: Using the Periodic Table as a Predictive Tool

The second part of the driving question asks whether you can predict an element's behaviour from its position alone. Using what you know about periodic trends, predict and justify the	The periodic table is not just an organisational chart — it is a predictive model. Once you understand the trends in electron configuration and ionisation energy, you can make reliable predictions about elements you have never tested. Potassium sits directly below sodium in Group I. It has one valence electron like sodium, but it is in Period 4, meaning that electron is in the fourth shell, shielded by three full inner shells. Its ionisation energy (419 kJ/mol) is lower than sodium's (496 kJ/mol), so potassium loses its valence electron even more easily. A water engineer in Kisumu could predict with confidence that potassium will react more violently with water than sodium — and the demonstration confirmed this. Calcium sits below magnesium in Group II. It has two

reactivity of (a) potassium (Group I, Period 4) compared to sodium, (b) calcium (Group II, Period 4) compared to magnesium, and (c) neon (Group 18, Period 2) compared to argon. Explain how a scientist or water engineer in Kenya could use the periodic table — without any experiment — to make reliable predictions about chemical behaviour.	valence electrons in the fourth shell, more shielded and further from the nucleus than magnesium's. Calcium reacts noticeably with cold water (producing calcium hydroxide and hydrogen gas), whereas magnesium barely reacts — a predictable and confirmed trend. Neon, like argon, is in Group 18 with a completely full outer shell. Its ionisation energy is even higher than argon's, and it is equally inert. A chemist studying gas mixtures at a Nairobi laboratory would correctly predict that neither neon nor argon will participate in any chemical reaction under normal conditions. The periodic table thus gives any scientist — or any Grade 10 student — a powerful tool: know an element's group and period, understand the trends, and you can predict its ionisation energy, reactivity, and chemical behaviour before a single experiment is run.

Section 5: Answering the Driving Question — Your Complete Scientific Argument	
Write a cohesive, well-structured paragraph (or set of paragraphs) that directly and fully answers the driving question: 'Why do elements that sit next to each other in the periodic table behave so differently, and can we use an element's position to predict exactly how it will behave?' Your answer must weave together all four previous sections into one scientific argument. Reference the Kisumu demonstration, name specific elements and their properties, and conclude with a statement about the power and purpose of the periodic table as a predictive scientific tool.	Elements that sit next to each other in the periodic table can behave dramatically differently because a single additional proton changes an atom's electron configuration — and it is electron configuration, not atomic number alone, that determines chemical behaviour. Sodium (atomic number 11) has one valence electron in its third shell, loosely held by a nucleus shielded by two inner shells of electrons. Its low ionisation energy means that valence electron is easily surrendered to water molecules, producing an explosive reaction that the Kisumu water officer demonstrated with breathtaking clarity: a fizzing, flaming piece of sodium skimming across a beaker of Lake Victoria water, leaving a strongly alkaline sodium hydroxide solution behind. Magnesium, with just one extra proton (atomic number 12), has two valence electrons and a higher effective nuclear charge — its ionisation energies are significantly higher, and cold water cannot supply enough energy to strip away its electrons, so it barely reacts. Aluminium, one step further, adds yet more nuclear charge, and reacts not at all with cold water. Argon, with a complete outer shell of eight electrons, has no chemical motivation to gain or lose electrons whatsoever — its ionisation energy is among the highest of all elements, explaining why it sits sealed and inert in the glass ampoule used to protect reactive sodium during transport. The reverse trend down Group I — where potassium is more reactive than sodium, which is more reactive than lithium — is explained by increasing electron shielding and increasing atomic radius as new shells are added, lowering the effective nuclear charge on the outer electron and dropping the ionisation energy with each period. Together, these trends — across periods and down groups — show that the periodic table is far more than a reference chart. It is a predictive scientific model. By knowing an element's group (which tells you its number of valence electrons) and its period (which tells you the size of its atom, the degree of shielding, and therefore its ionisation energy), you can predict with confidence how reactive it will be, what compounds it will form, and how it will behave in a test tube — before you ever open one. The water purification officer at Kisumu did not need to experiment with potassium to know it would be more dangerous than sodium; the periodic table told the whole story in advance.

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Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Scientific Accuracy of Core Concepts	Electron configuration, ionisation energy, shielding, effective nuclear charge, and valence electrons are all defined and applied correctly and precisely. The	Most core concepts are correctly applied. Minor inaccuracies (e.g., imprecise description of shielding) do not undermine the overall argument.	One or more key concepts (e.g., ionisation energy, shielding) are missing or significantly misapplied. The explanation of the sodium–water reaction

	sodium–water reaction products (NaOH, H_2, heat) are correctly identified. No scientific errors are present.	Reaction products are mostly correct.	is incomplete or contains errors.
Connection to the Anchoring Phenomenon	Every section connects explicitly and specifically to the Kisumu demonstration. Named elements (Na, Mg, Al, Ar, K) from the demonstration are referenced with accurate particle-level explanations for their observed behaviour. The explanation tells a coherent story from the phenomenon to the science.	The phenomenon is referenced in most sections. At least three elements from the demonstration are explained using particle-level reasoning. The connection is clear but may lack detail in one or two places.	The phenomenon is mentioned only briefly or superficially. Fewer than two elements are explained at the particle level. The scientific argument is not clearly linked to what was observed.
Explanation of Periodic Trends (Across and Down)	Both trends are explained fully and accurately: ionisation energy increases left to right across Period 3 (with clear reasoning from nuclear charge and shielding) and decreases down Group I (with clear reasoning from added electron shells and increased shielding). Both trends are connected to observed reactivity differences.	Both trends are identified and mostly correctly explained. The reasoning for one trend (across or down) may be less detailed or precise than the other.	Only one trend is addressed, or both trends are present but the reasoning is vague or incorrect (e.g., stating only that 'atoms get bigger' without explaining why that affects reactivity).
Use of the Periodic Table as a Predictive Tool	The student makes at least two specific, well-justified predictions about elements not demonstrated (e.g., potassium, calcium, neon) using group and period reasoning. The student articulates clearly how and why the periodic table allows predictions before experimentation.	At least one specific prediction is made and justified using periodic trends. The student acknowledges the predictive power of the table but may not fully explain the underlying reasoning.	Predictions are absent, vague, or not linked to the structure of the periodic table. The student does not demonstrate understanding of the periodic table as a predictive tool.
Quality of Scientific Argument and Communication	The Final Explanation is a cohesive, well-structured scientific argument. Ideas flow logically from atomic structure through trends to the phenomenon and to prediction. Scientific vocabulary is used correctly and consistently. The driving question is answered directly, completely, and in the student's own words.	The argument is mostly coherent and addresses the driving question. Scientific vocabulary is mostly used correctly. Some sections may feel disconnected or the conclusion may lack full development.	The response reads as a list of disconnected facts rather than a scientific argument. The driving question is only partially answered. Scientific vocabulary is limited, missing, or misused.